

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

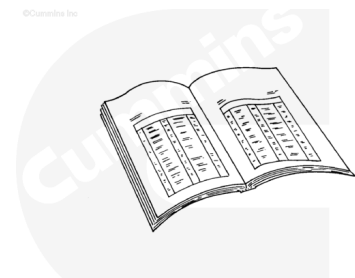
To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-037-tr.html>)
- Remove the oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html>)

©Cummins Inc.



ck800wa

Remove

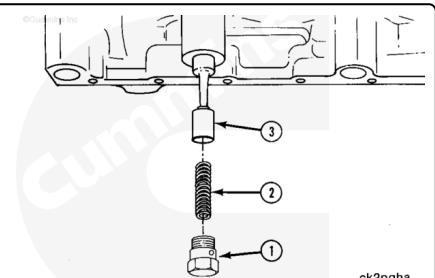
⚠ WARNING ⚠

Carefully remove the retainer plug. The pressure regulator spring is under compression. Personal injury can result. Wear face and eye protection.

Remove the retainer plug (1), regulator spring (2), and regulator plunger (3).

Remove the orifice from the plunger.

©Cummins Inc.



ck2pgha

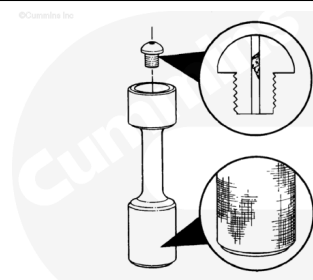
Inspect for Reuse

Inspect the regulator plunger for scratches or scoring. If scratches are deep enough to be felt with a fingernail, the plunger **must** be replaced.

Check the orifice in the plunger to make sure it is **not** plugged. If the orifice is plugged, clean it before replacing.

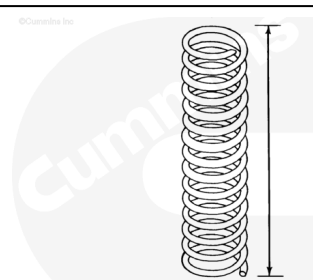
The orifice is threaded into the plunger.

Torque Value: 1 n•m [6 in-lb]



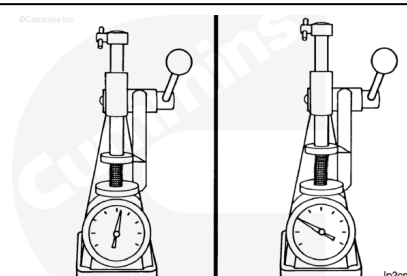
Measure the free length of the regulator spring.

Regulator Spring Free Length		
mm		in
84.1	NOM	3.31



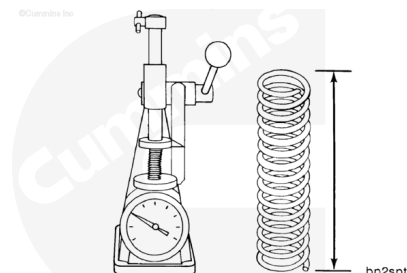
Use a spring compression tester to determine if the spring is defective.

Replace the spring with a new spring if it is defective.



Main Oil Pressure Regulator Spring Load at 48.3 mm
[1.90 in]

n		lbf
91.1	MIN	20.50
94.7	MAX	21.30

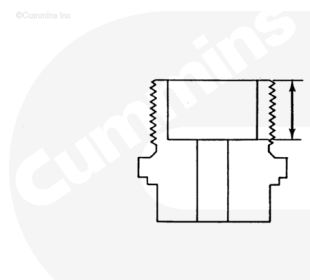


Measure the retainer plug to determine if it is defective. If it does **not** meet the specification, replace it with a new one.

Main Oil Pressure Regulator Retainer Plug		
mm		in
11.1	MIN	0.44

Main Oil Pressure Regulator Retainer Plug

mm		in
13.4	MAX	0.53

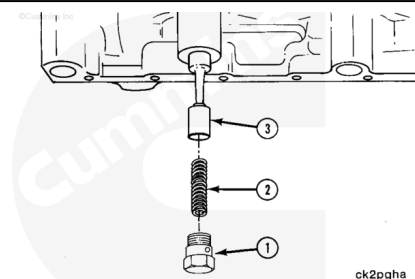


bp2rta

Install

⚠ WARNING ⚠

The regulator spring must be compressed to install the retainer plug. Personal injury can result. Wear face and eye protection.



ck2pgha

Note : A counterbore plunger and orifice is unique to engines with a viscosity sensor. Low oil pressure will result if a solid plunger is used with a viscosity sensor.

Install the regulator plunger (3) and regulator spring (2).

Install and tighten the retainer plug (1).

Torque Value:

Rear Sump Oil Pan 75 n•m [55 ft-lb]

Torque Value:

Front Sump Oil Pan 122 n•m [90 ft-lb]

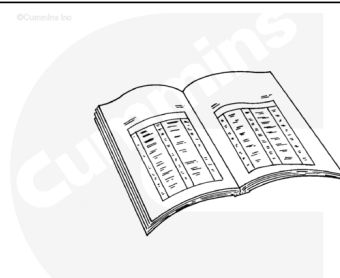
Torque Value:

Center Sump Oil Pan 75 n•m [55 ft-lb]

Finishing Steps

⚠ CAUTION ⚠

Engine oil pressure must be indicated on the gauge within 15 seconds after starting. If oil pressure is not registered within 15 seconds, shut off the engine immediately to avoid engine damage. Confirm the correct oil level in the oil pan.



ck800wa

- Install the oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html>)
- Fill the engine with clean 15W-40 engine oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-037-tr.html>)
- Operate the engine at idle for 3 minutes and check for leaks.

Last Modified: 20-Sep-2013
